A Technical Introduction to Recommender system

Section 1

Time and Seasonality: what specify time and date or a season that a user interacts with the platform User Navigation Patterns: sequence of pages visited, time spent on different parts of a website, mouse movement, etc. External Social Trends: information from outer social media

Section 2

Hybrid recommendations approaches Most recommender systems now use a hybrid approach, combining collaborative filtering, content-based filtering, and other approaches. E-commerce platforms frequently use hybrid approaches to overcome problems like the cold start problem, where a new user has no history for collaborative filtering to analyze. There is no reason why several different techniques of the same type could not be hybridized. Hybrid approaches can be implemented in several ways: by making content-based and collaborative-based predictions separately and then combining them; by adding content-based capabilities to a collaborative-based approach (and vice versa); or by unifying the approaches into one model. Several studies that empirically compared the performance of the hybrid with the pure collaborative and content-based methods and demonstrated that the hybrid methods can provide more accurate recommendations than pure approaches. These methods can also be used to overcome some of the common problems in recommender systems such as cold start and the sparsity problem, as well as the knowledge engineering bottleneck in knowledge-based approaches. Netflix is a good example of the use of hybrid recommender systems. The website makes recommendations by comparing the watching and searching habits of similar users (i.e., collaborative filtering) as well as by offering movies that share characteristics with films that a user has rated highly (content-based filtering). Some hybridization techniques include:

Section 3

One of the events that energized research in recommender systems was the Netflix Prize. From 2006 to 2009, Netflix sponsored a competition, offering a grand prize of $1,000,000 to the team that could take an offered dataset of over 100 million movie ratings and return recommendations that were 10% more accurate than those offered by the company's existing recommender system. This competition energized the search for new and more accurate algorithms. On 21 September 2009, the grand prize of US$1,000,000 was given to the BellKor's Pragmatic Chaos team using tiebreaking rules. The most accurate algorithm in 2007 used an ensemble method of 107 different algorithmic approaches, blended into a single prediction. As stated by the winners, Bell et al.:

Section 4

Asking a user to rate an item on a sliding scale. Asking a user to search. Asking a user to rank a collection of items from favorite to least favorite. Presenting two items to a user and asking him/her to choose the better one of them. Asking a user to create a list of items that he/she likes (see Rocchio classification or other similar techniques). Examples of implicit data collection include the following:

Section 5

In contrast to an engagement-based ranking system employed by social media and other digital platforms, a bridging-based ranking optimizes for content that is unifying instead of polarizing. Examples include Polis and Remesh which have been used around the world to help find more consensus around specific political issues. Twitter has also used this approach for managing its community notes, which YouTube planned to pilot in 2024. Aviv Ovadya also argues for implementing bridging-based algorithms in major platforms by empowering deliberative groups that are representative of the platform's users to control the design and implementation of the algorithm.

Section 6

A model of the user's preference. A history of the user's interaction with the recommender system. Basically, these methods use an item profile (i.e., a set of discrete attributes and features) characterizing the item within the system. To abstract the features of the items in the system, an item presentation algorithm is applied. A widely used algorithm is the tf–idf representation (also called vector space representation). The system creates a content-based profile of users based on a weighted vector of item features. The weights denote the importance of each feature to the user and can be computed from individually rated content vectors using a variety of techniques. Simple approaches use the average values of the rated item vector while other sophisticated methods use machine learning techniques such as Bayesian Classifiers, cluster analysis, decision trees, and artificial neural networks in order to estimate the probability that the user is going to like the item. A key issue with content-based filtering is whether the system can learn user preferences from users' actions regarding one content source and use them across other content types. When the system is limited to recommending content of the same type as the user is already using, the value from the recommendation system is significantly less than when other content types from other services can be recommended. For example, recommending news articles based on news browsing is useful. Still, it would be much more useful when music, videos, products, discussions, etc., from different services, can be recommended based on news browsing. To overcome this, most content-based recommender systems now use some form of the hybrid system. Content-based recommender systems can also include opinion-based recommender systems. In some cases, users are allowed to leave text reviews or feedback on the items. These user-generated texts are implicit data for the recommender system because they are potentially rich resources of both feature/aspects of the item and users' evaluation/sentiment to the item. Features extracted from the user-generated reviews are improved metadata of items, because as they also reflect aspects of the item like metadata, extracted features are widely concerned by the users. Sentiments extracted from the reviews can be seen as users' rating scores on the corresponding features. Popular approaches of opinion-based recommender system utilize various techniques including text mining, information retrieval, sentiment analysis (see also Multimodal sentiment analysis) and deep learning.

Section 7

Further reading Books Kim Falk (d 2019), Practical Recommender Systems, Manning Publications, ISBN 9781617292705 Bharat Bhasker; K. Srikumar (2010). Recommender Systems in E-Commerce. CUP. ISBN 978-0-07-068067-8. Archived from the original on September 1, 2010. Jannach, Dietmar; Markus Zanker; Alexander Felfernig; Gerhard Friedrich (2010). Recommender Systems: An Introduction. CUP. ISBN 978-0-521-49336-9. Archived from the original on August 31, 2015. Seaver, Nick (2022). Computing Taste: Algorithms and the Makers of Music Recommendation. University of Chicago Press. Scientific articles Robert M. Bell; Jim Bennett; Yehuda Koren & Chris Volinsky (May 2009). "The Million Dollar Programming Prize". IEEE Spectrum. Archived from the original on May 11, 2009. Retrieved December 10, 2018. Prem Melville, Raymond J. Mooney, and Ramadass Nagarajan. (2002) Content-Boosted Collaborative Filtering for Improved Recommendations. Proceedings of the Eighteenth National Conference on Artificial Intelligence (AAAI-2002), pp. 187–192, Edmonton, Canada, July 2002.